Thermal harmonic oscillator: the computation

Fourier modes, with a circle discretization as a check

Take $m, \omega, \beta > 0$, with $\beta = 1/(k_B T)$ and $L = \beta\hbar$. Our starting expression is

$$Z = \int_{x(L)=x(0)} \mathcal{D}x\, e^{-S_E[x]/\hbar}, \qquad S_E[x] = \frac{m}{2}\int_0^L \mathrm{d}\tau\, [\dot{x}^{\,2} + \omega^2 x^2]. \tag{1}$$

1. Expand the path and compute the action

First retain only M positive frequencies:

$$x_M(\tau) = \bar{x} + \sum_{n=1}^{M} [a_n \cos(\nu_n \tau) + b_n \sin(\nu_n \tau)], \qquad \nu_n = \frac{2\pi n}{L}. \tag{2}$$

All coefficients are real. The average $\bar{x} = L^{-1}\int_0^L x_M \mathrm{d}\tau$ is an integration variable; it is not fixed to zero or to the endpoint. For positive integers n, r, orthogonality reads

$$\int_0^L \cos(\nu_n\tau)\cos(\nu_r\tau)\mathrm{d}\tau = \int_0^L \sin(\nu_n\tau)\sin(\nu_r\tau)\mathrm{d}\tau = \frac{L}{2}\delta_{nr},$$
$$\int_0^L \cos(\nu_n\tau)\sin(\nu_r\tau)\mathrm{d}\tau = 0, \qquad \int_0^L \cos(\nu_n\tau)\mathrm{d}\tau = \int_0^L \sin(\nu_n\tau)\mathrm{d}\tau = 0.$$

Thus all cross terms vanish when we square and integrate:

$$\int_0^L x_M^2 \mathrm{d}\tau = L\bar{x}^2 + \frac{L}{2}\sum_{n=1}^{M}(a_n^2 + b_n^2), \tag{3}$$

$$\dot{x}_M = \sum_{n=1}^{M} \nu_n[-a_n \sin(\nu_n\tau) + b_n \cos(\nu_n\tau)], \tag{4}$$

$$\int_0^L \dot{x}_M^2 \mathrm{d}\tau = \frac{L}{2}\sum_{n=1}^{M}\nu_n^2(a_n^2 + b_n^2). \tag{5}$$

Insert these expressions in the action:

$$\boxed{\frac{S_E[x_M]}{\hbar} = \frac{mL\omega^2}{2\hbar}\bar{x}^2 + \sum_{n=1}^{M}\frac{mL}{4\hbar}(\nu_n^2 + \omega^2)(a_n^2 + b_n^2).} \tag{6}$$

The constant mode has no kinetic term. Each $n > 0$ has *two* real coordinates, a cosine coefficient and a sine coefficient. We will integrate all $2M + 1$ coordinates, then let $M \to \infty$.

2. Fix the measure from phase space, without a propagator

The normalization convention is the quantum trace with $H = p^2/(2m) + m\omega^2 x^2/2$, with no added constant in H. Here is a direct way to fix its measure. Set $\epsilon = L/N$ and use

$$\langle x|p\rangle = \frac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}}, \qquad \int \mathrm{d}x\, |x\rangle\langle x| = 1, \qquad \int \mathrm{d}p\, |p\rangle\langle p| = 1.$$

Insert these resolutions into the Trotter product for the trace, placing one position and one momentum integral on each step. With $x_N = x_0$,

$$Z = \lim_{N\to\infty} \int \prod_{j=0}^{N-1} \frac{\mathrm{d}x_j\, \mathrm{d}p_j}{2\pi\hbar} \exp\left[\sum_j \left\{ \frac{ip_j(x_{j+1} - x_j)}{\hbar} - \frac{\epsilon}{\hbar}\left(\frac{p_j^2}{2m} + \frac{m\omega^2 x_j^2}{2}\right)\right\}\right]. \tag{7}$$

This uses only the position–momentum overlap, not a known propagator. The factors $1/(2\pi\hbar)$ come from the two overlaps on each step.

Evaluate a momentum integral explicitly. For $a > 0$, define $F(b) = \int_{\mathbb{R}} e^{-ap^2+ibp}\mathrm{d}p$. Integration by parts gives

$$0 = \int \frac{\mathrm{d}}{\mathrm{d}p} e^{-ap^2+ibp}\mathrm{d}p = -2a\int p e^{-ap^2+ibp}\mathrm{d}p + ibF(b), \qquad F'(b) = -\frac{b}{2a}F(b).$$

Since $F(0) = \sqrt{\pi/a}$, solving this differential equation yields $F(b) = \sqrt{\pi/a}\, e^{-b^2/(4a)}$. Take $a = \epsilon/(2m\hbar)$ and $b = (x_{j+1} - x_j)/\hbar$:

$$\int \frac{\mathrm{d}p_j}{2\pi\hbar} e^{-\epsilon p_j^2/(2m\hbar) + ip_j(x_{j+1}-x_j)/\hbar} = \sqrt{\frac{m}{2\pi\hbar\epsilon}} e^{-m(x_{j+1}-x_j)^2/(2\hbar\epsilon)}. \tag{8}$$

Thus the absolutely normalized position measure is the limit of

$$\boxed{\prod_{j=0}^{N-1} \sqrt{\frac{m}{2\pi\hbar\epsilon}}\, \mathrm{d}x_j.} \tag{9}$$

This calculation supplies the measure needed here and in Section 8. There is no need to introduce a propagator or a diagonal kernel.

What zeta regularization will do. It assigns a value to the infinite Gaussian determinant in this canonical normalization convention. It cannot determine an arbitrary extra factor inserted by hand into the measure. Such a factor would change the trace; a factor $e^{-\beta E_*}$ would in particular change H to $H + E_*$.

3. Fourier Gaussians as a spectral determinant

Use an orthonormal real basis on the circle:

$$e_0 = \frac{1}{\sqrt{L}}, \qquad e_{n,c} = \sqrt{\frac{2}{L}}\cos(\nu_n \tau), \qquad e_{n,s} = \sqrt{\frac{2}{L}}\sin(\nu_n \tau).$$

Writing $x = Q_0 e_0 + \sum_{n\geq 1}(Q_{n,c}e_{n,c} + Q_{n,s}e_{n,s})$ gives

$$Q_0 = \sqrt{L}\,\bar{x}, \qquad Q_{n,c} = \sqrt{L/2}\,a_n, \qquad Q_{n,s} = \sqrt{L/2}\,b_n.$$

The action and its eigenvalues are now

$$\frac{S_E}{\hbar} = \frac{m}{2\hbar}\left[\omega^2 Q_0^2 + \sum_{n\geq 1}(\nu_n^2 + \omega^2)(Q_{n,c}^2 + Q_{n,s}^2)\right], \quad \lambda_0 = \omega^2, \quad \lambda_{n,c} = \lambda_{n,s} = \nu_n^2 + \omega^2. \tag{10}$$

Introduce a reference frequency $\mu > 0$ so that the spectral eigenvalues λ_α/μ^2 are dimensionless. For one real coordinate,

$$\int_{\mathbb{R}} \mu\sqrt{\frac{m}{2\pi\hbar}}\mathrm{d}Q_\alpha\, e^{-m\lambda_\alpha Q_\alpha^2/(2\hbar)} = \frac{\mu}{\sqrt{\lambda_\alpha}}. \tag{11}$$

This is the usual Gaussian convention $\mathrm{d}y/\sqrt{2\pi}$, with $y = \mu\sqrt{m/\hbar}\,Q_\alpha$. Its infinite product requires a prescription; it is not a literal change of variables in a flat infinite-dimensional volume. In the canonical convention of Section 2 we use the zeta determinant below; Sections 8–9 independently verify its normalization using finite integrals.

For finitely many eigenvalues,

$$\zeta_A(s) = \sum_\alpha (\lambda_\alpha/\mu^2)^{-s} \quad \Longrightarrow \quad \zeta_A'(0) = -\sum_\alpha \log(\lambda_\alpha/\mu^2).$$

The zeta prescription keeps this relation after analytic continuation:

$$\boxed{\det{}_\zeta(A/\mu^2) = e^{-\zeta_A'(0)}, \qquad Z_\zeta = [\det{}_\zeta(A/\mu^2)]^{-1/2} = e^{\zeta_A'(0)/2},} \qquad A = -\partial_\tau^2 + \omega^2. \tag{12}$$

For our circle,

$$\zeta_A(s) = \left(\frac{\omega}{\mu}\right)^{-2s} + 2\sum_{n=1}^{\infty}\left(\frac{\nu_n^2 + \omega^2}{\mu^2}\right)^{-s}. \tag{13}$$

The series initially converges for $\mathrm{Re}\, s > 1/2$. The coefficient 2 counts the sine and cosine separately. We next continue the expression to $s = 0$ and evaluate its derivative.

4. Derive the two Riemann-zeta values we need

For $\operatorname{Re} t > 1$, define $\zeta_R(t) = \sum_{n=1}^{\infty} n^{-t}$. We need its continuation to $t = 0$. Euler–Maclaurin summation (integral plus endpoint correction) gives

$$\zeta_R(t) = \lim_{K\to\infty} \left[\sum_{n=1}^{K} n^{-t} - \frac{K^{1-t}}{1-t} - \frac{1}{2}K^{-t} \right]. \tag{14}$$

This limit is valid for $\operatorname{Re} t > -1$, except for the pole at $t = 1$. For $\operatorname{Re} t > 1$ it agrees with the original series. To see the continuation, the tail's integral is $K^{1-t}/(t-1)$ and its endpoint correction is $-K^{-t}/2$; the next remainder is $O(K^{-\operatorname{Re} t-1})$, locally uniformly in t. The same formula therefore defines an analytic function near zero, where the limit can also be differentiated.

Step 4a: the value at zero. Substituting $t = 0$ in the bracket gives

$$\boxed{\zeta_R(0) = \lim_{K\to\infty} [K - K - \tfrac{1}{2}] = -\tfrac{1}{2}.} \tag{15}$$

This is an analytically continued value, not the sum of the positive numbers $1 + 1 + 1 + \cdots$ in the ordinary sense.

Step 4b: the derivative at zero. Differentiate each term in (14):

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} \sum_{n=1}^{K} n^{-t} \bigg|_{t=0} &= -\sum_{n=1}^{K} \log n = -\log(K!), \\
\frac{\mathrm{d}}{\mathrm{d}t} \left[-\frac{K^{1-t}}{1-t} \right] \bigg|_{t=0} &= K \log K - K, \\
\frac{\mathrm{d}}{\mathrm{d}t} \left[-\tfrac{1}{2}K^{-t} \right] \bigg|_{t=0} &= \tfrac{1}{2} \log K.
\end{aligned}$$

Hence

$$\zeta_R'(0) = \lim_{K\to\infty} [-\log(K!) + K \log K - K + \tfrac{1}{2} \log K].$$

Use Stirling's formula with its constant displayed:

$$\log(K!) = (K + \tfrac{1}{2}) \log K - K + \tfrac{1}{2} \log(2\pi) + O(K^{-1}).$$

All the K-dependent terms cancel, leaving

$$\boxed{\zeta_R'(0) = -\frac{1}{2} \log(2\pi).} \tag{16}$$

These are the only special zeta values needed in the computation.

5. Evaluate the regularized Fourier determinant

First isolate one copy of the positive-frequency free spectrum:

$$F(s) = \sum_{n=1}^{\infty} (\nu_n^2/\mu^2)^{-s} = \left(\frac{L\mu}{2\pi}\right)^{2s} \zeta_R(2s).$$

Using the values just derived,

$$F(0) = -\tfrac{1}{2}, \tag{17}$$

$$F'(0) = 2\log\left(\frac{L\mu}{2\pi}\right)\zeta_R(0) + 2\zeta_R'(0) = -\log(L\mu). \tag{18}$$

Thus $\prod_{n\geq 1}^{\zeta}(\nu_n^2/\mu^2) = e^{-F'(0)} = L\mu$. Counting *both* real coordinates gives the nonzero-mode determinant $e^{-2F'(0)} = (L\mu)^2$. These two products must not be confused.

To include the potential, write

$$\zeta_A(s) = (\omega/\mu)^{-2s} + 2F(s) + 2D(s), \tag{19}$$

$$D(s) = \sum_{n=1}^{\infty}\left[\left(\frac{\nu_n^2+\omega^2}{\mu^2}\right)^{-s} - \left(\frac{\nu_n^2}{\mu^2}\right)^{-s}\right]. \tag{20}$$

For s near zero each difference is $O(n^{-2\,\mathrm{Re}\,s-2})$. The series is analytic there and can be differentiated term by term:

$$D(0) = 0, \qquad D'(0) = -\sum_{n=1}^{\infty}\log\left(1+\frac{\omega^2}{\nu_n^2}\right).$$

We can now evaluate the full spectral zeta function at zero:

$$\zeta_A(0) = 1 + 2(-\tfrac{1}{2}) = 0, \tag{21}$$

$$\zeta_A'(0) = -2\log(\omega/\mu) - 2\log(L\mu) - 2\sum_{n=1}^{\infty}\log\left(1+\frac{\omega^2}{\nu_n^2}\right) \tag{22}$$

$$= -2\log(L\omega) - 2\sum_{n=1}^{\infty}\log\left(1+\frac{\omega^2}{\nu_n^2}\right). \tag{23}$$

The arbitrary reference frequency μ cancels. Exponentiating half the last expression gives

$$\boxed{Z_\zeta = \frac{1}{L\omega}\prod_{n=1}^{\infty}\left(1+\frac{\omega^2}{\nu_n^2}\right)^{-1}.} \tag{24}$$

In particular, the prefactor $1/(L\omega)$ was not guessed from the oscillator spectrum. The zero mode contributes μ/ω and the regularized free nonzero-mode Gaussian product contributes $1/(L\mu)$.

6. Recover the normalized measure and finite Gaussians

Here is the explicit link to a calculation with a finite Fourier cutoff. The raw Gaussian measure in the orthonormal coordinates is

$$\mathrm{d}\rho_M = \mu\sqrt{\frac{m}{2\pi\hbar}}\mathrm{d}Q_0 \prod_{n=1}^{M}\left[\frac{\mu^2 m}{2\pi\hbar}\mathrm{d}Q_{n,c}\,\mathrm{d}Q_{n,s}\right].$$

For each free nonconstant pair its integral is μ^2/ν_n^2. To replace this vanishing cutoff product by its nonzero zeta value $1/(L\mu)$, normalize each pair to one, then put that value back once:

$$\mathcal{D}_M x = \frac{1}{L\mu}\,\mathrm{d}\rho_M \prod_{n=1}^{M}\frac{\nu_n^2}{\mu^2}. \tag{25}$$

Indeed, inserting a pair's kinetic weight gives

$$\frac{\mu^2 m}{2\pi\hbar}\frac{\nu_n^2}{\mu^2}\int \mathrm{d}Q_{n,c}\,\mathrm{d}Q_{n,s}\,e^{-m\nu_n^2(Q_{n,c}^2+Q_{n,s}^2)/(2\hbar)} = 1.$$

Now use the coefficient Jacobians $\mathrm{d}Q_0 = \sqrt{L}\,\mathrm{d}\bar{x}$ and $\mathrm{d}Q_{n,c}\mathrm{d}Q_{n,s} = (L/2)\mathrm{d}a_n\mathrm{d}b_n$:

$$\boxed{\mathcal{D}_M x = \sqrt{\frac{m}{2\pi\hbar L}}\mathrm{d}\bar{x}\prod_{n=1}^{M}\left[\frac{mL\nu_n^2}{4\pi\hbar}\mathrm{d}a_n\,\mathrm{d}b_n\right].} \tag{26}$$

This derives the normalization factors without any diagonal kernel. Use $\int \mathrm{d}q\,e^{-Aq^2} = \sqrt{\pi/A}$ for $A > 0$ to do the remaining integrals:

$$I_0 = \sqrt{\frac{m}{2\pi\hbar L}}\int \mathrm{d}\bar{x}\,e^{-mL\omega^2\bar{x}^2/(2\hbar)} = \sqrt{\frac{m}{2\pi\hbar L}}\sqrt{\frac{2\pi\hbar}{mL\omega^2}} = \frac{1}{L\omega}, \tag{27}$$

$$I_n = \frac{mL\nu_n^2}{4\pi\hbar}\int \mathrm{d}a_n\mathrm{d}b_n\,e^{-mL(\nu_n^2+\omega^2)(a_n^2+b_n^2)/(4\hbar)} \tag{28}$$

$$= \frac{mL\nu_n^2}{4\pi\hbar}\left[\sqrt{\frac{4\pi\hbar}{mL(\nu_n^2+\omega^2)}}\right]^2 = \frac{\nu_n^2}{\nu_n^2+\omega^2}. \tag{29}$$

Therefore $Z_M = I_0\prod_{n=1}^{M} I_n$. Put $z = L\omega/2$. Since $0 \le \log(1+z^2/(\pi^2 n^2)) \le z^2/(\pi^2 n^2)$, the remaining logarithmic series converges ordinarily, giving

$$Z = \lim_{M\to\infty} Z_M = \frac{1}{2z}\left[\prod_{n=1}^{\infty}\left(1+\frac{z^2}{\pi^2 n^2}\right)\right]^{-1}. \tag{30}$$

No extra product over negative frequencies is needed: each positive n already includes two real coordinates. The mass and μ have both canceled.

7. Evaluate the product, without quoting a product identity

We need $P(z) = \prod_{n\geq 1}(1 + z^2/(\pi^2 n^2))$. The following derivation uses just one more Fourier series. For fixed $z > 0$, expand the even function $f(t) = \cosh(zt)$ on $[-1, 1]$:

$$\cosh(zt) = \frac{d_0}{2} + \sum_{n=1}^{\infty} d_n \cos(n\pi t).$$

The coefficient convention is $d_n = \int_{-1}^{1} f(t)\cos(n\pi t)\mathrm{d}t$. First $d_0 = 2\sinh z/z$. For $n \geq 1$, the antiderivative

$$\int \cosh(zt)\cos(n\pi t)\mathrm{d}t = \frac{z\sinh(zt)\cos(n\pi t) + n\pi\cosh(zt)\sin(n\pi t)}{z^2 + \pi^2 n^2}$$

gives, by evaluating at $t = \pm 1$,

$$d_n = \frac{2z\sinh z\,(-1)^n}{z^2 + \pi^2 n^2}.$$

The periodic extension is continuous at the endpoints, and these coefficients are $O(n^{-2})$. Thus the Fourier series converges to $\cosh z$ at $t = 1$. Using $\cos(n\pi) = (-1)^n$ there gives

$$\cosh z = \frac{\sinh z}{z} + 2z\sinh z \sum_{n=1}^{\infty} \frac{1}{z^2 + \pi^2 n^2}, \tag{31}$$

$$\coth z - \frac{1}{z} = \sum_{n=1}^{\infty} \frac{2z}{z^2 + \pi^2 n^2}. \tag{32}$$

Integrate from 0 to z. The apparent singularity on the left is removable; $\sinh s/s \to 1$ as $s \to 0$. On the right, $0 \leq 2s/(s^2 + \pi^2 n^2) \leq 2z/(\pi^2 n^2)$ for $0 \leq s \leq z$. This summable bound gives uniform convergence and justifies integrating term by term. Therefore

$$\log\frac{\sinh z}{z} = \sum_{n=1}^{\infty} \int_0^z \frac{2s\,\mathrm{d}s}{s^2 + \pi^2 n^2} \tag{33}$$

$$= \sum_{n=1}^{\infty} \log\left(1 + \frac{z^2}{\pi^2 n^2}\right) = \log P(z). \tag{34}$$

Exponentiating and substituting into (30), we obtain

$$P(z) = \frac{\sinh z}{z}, \qquad \boxed{Z = \frac{1}{2z}\frac{z}{\sinh z} = \frac{1}{2\sinh(\beta\hbar\omega/2)}}. \tag{35}$$

Finally, with $b = \beta\hbar\omega$, rearrange the result:

$$Z = \frac{e^{-b/2}}{1 - e^{-b}} = \sum_{r=0}^{\infty} e^{-b(r+1/2)}.$$

This recovers the partition function for energies $E_r = \hbar\omega(r + 1/2)$; the spectrum was not used to fix the path-integral normalization.

8. Discretize the circle: an ordinary Gaussian integral

Divide the circle into $N \geq 3$ steps, $\epsilon = L/N$, and set $x_N = x_0$. In this section x_j denotes the position at site j; x_0 is a site value, not the Fourier average $\bar{x}$. The Trotter product gives

$$Z = \lim_{N\to\infty} Z_N, \qquad Z_N = \left(\frac{m}{2\pi\hbar\epsilon}\right)^{N/2} \int_{\mathbb{R}^N} \prod_{j=0}^{N-1} \mathrm{d}x_j \, \exp\left[-\frac{m}{2\hbar\epsilon}\sum_{j=0}^{N-1}\left((x_{j+1}-x_j)^2 + (\epsilon\omega)^2 x_j^2\right)\right]. \tag{36}$$

Specifically, Z_N equals the trace of $(e^{-\epsilon p^2/(2m\hbar)} e^{-\epsilon m\omega^2 x^2/(2\hbar)})^N$. This trace approaches Z as $N \to \infty$; at finite N it is an approximation.

Put $u = \epsilon\omega$ and let A be the real symmetric matrix defined by

$$x^{\mathsf{T}} A x = \sum_j [(x_{j+1}-x_j)^2 + u^2 x_j^2], \qquad (Ax)_j = (2+u^2)x_j - x_{j+1} - x_{j-1},$$

with indices taken modulo N. Its eigenvectors are discrete plane waves:

$$v_j^{(k)} = N^{-1/2} e^{2\pi i k j/N}, \qquad \lambda_k = 2 + u^2 - 2\cos\frac{2\pi k}{N} = u^2 + 4\sin^2\frac{\pi k}{N}, \quad 0 \leq k < N.$$

Complex waves are only a convenient way to find the eigenvalues. The integral still has exactly N real coordinates; a real orthogonal change of variables has Jacobian of absolute value one.

Apply the N-dimensional Gaussian formula:

$$\int \mathrm{d}^N x \, \exp\left[-\frac{m}{2\hbar\epsilon} x^{\mathsf{T}} A x\right] = \left(\frac{2\pi\hbar\epsilon}{m}\right)^{N/2} (\det A)^{-1/2}, \tag{37}$$

$$\boxed{Z_N = (\det A)^{-1/2}} = \prod_{k=0}^{N-1} \left[u^2 + 4\sin^2\frac{\pi k}{N}\right]^{-1/2}. \tag{38}$$

All prefactors have now canceled explicitly. For $\omega > 0$, every λ_k is positive, so the integral is finite and the square root is positive.

Connection to Fourier modes. The eigenvalues of the discretized fluctuation operator are

$$\epsilon^{-2}\lambda_k = \omega^2 + \left[\frac{2}{\epsilon}\sin\frac{\pi k}{N}\right]^2.$$

For fixed positive k, the second term tends to $(2\pi k/L)^2$. The modes near N supply the negative frequencies. We evaluate the full finite product next, rather than exchanging a limit with a growing product.

9. Evaluate the finite product, then take the limit

Introduce $\eta > 0$ by

$$u = 2\sinh(\eta/2), \qquad \eta = 2\,\mathrm{arsinh}(u/2).$$

Writing $\theta_k = 2\pi k/N$, each eigenvalue becomes

$$\lambda_k = 2\cosh\eta - 2\cos\theta_k = e^{\eta}(1 - e^{-\eta}e^{i\theta_k})(1 - e^{-\eta}e^{-i\theta_k}).$$

The roots-of-unity identity

$$\prod_{k=0}^{N-1}(1 - re^{2\pi ik/N}) = 1 - r^N$$

follows from $z^N - 1 = \prod_k (z - e^{2\pi ik/N})$. Use it twice, with $r = e^{-\eta}$:

$$\det A = e^{N\eta}(1 - e^{-N\eta})^2 = 4\sinh^2(N\eta/2), \tag{39}$$

$$Z_N = \boxed{\frac{1}{2\sinh\left[N\,\mathrm{arsinh}\left(\frac{L\omega}{2N}\right)\right]}}. \tag{40}$$

This expression is exact for the discretized integral (36). Since $\mathrm{arsinh}\, y = y + O(y^3)$ as $y \to 0$,

$$N\,\mathrm{arsinh}\left(\frac{L\omega}{2N}\right) = \frac{L\omega}{2} - \frac{(L\omega)^3}{48N^2} + O(N^{-4}) \quad \text{at fixed } L\omega.$$

Thus

$$\boxed{\lim_{N\to\infty} Z_N = \frac{1}{2\sinh(\beta\hbar\omega/2)}}. \tag{41}$$

The discretized calculation agrees with the Fourier calculation.

Checks to ask the students

High temperature: for $b = \beta\hbar\omega \ll 1$, $Z \sim 1/b$. This matches the classical phase-space integral with one state per phase-space area $2\pi\hbar$:

$$Z_{\mathrm{cl}} = \frac{1}{2\pi\hbar}\int \mathrm{d}p\,\mathrm{d}x\, e^{-\beta[p^2/(2m) + m\omega^2x^2/2]} = \frac{1}{\beta\hbar\omega}.$$

Low temperature: for $b \gg 1$, $Z \sim e^{-b/2}$, revealing the ground-state energy $\hbar\omega/2$. More generally, $-\partial_\beta \log Z = (\hbar\omega/2)\coth(b/2)$.

Removing the trap: as $\omega \to 0^+$, Z diverges. The constant mode is no longer confined; a free particle on $\mathbb{R}$ has an infinite spatial trace. This is an expected physical divergence.